

ACTIVITY 21 — FACILITATOR GUIDE**Root Cause Analysis with 5 Whys, Fishbone and Pareto**

TYPE Workshop	DURATION 60 minutes	MAPS TO LO3	TOPIC 5 - Monitor and Control the Project
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Purpose of this activity

By the end of this activity learners will be able to use Pareto analysis to find the vital few defect categories driving most of the pain.

What learners produce

pareto analysis, fishbone, 5whys — submitted to artifacts/.

Materials required

- Learner Worksheet (one per learner)
- Checklist (one per group)
- The Project Management Templates pack (see the Activities folder)
- Laptop with internet access for the live problem-solving tools
- The course slides for the framework and worked example

Set-up

Groups of 3 to 4. Distribute one worksheet per learner and one checklist per group; each group needs one laptop for the calculations.

Scenario / briefing material

Contoso Learning Pte Ltd is a training provider with 14,000 active learners. Its course registration portal is 9 years old. Registration takes a median of 11 minutes and 34% of attempts are abandoned. Learner communications consume 24 staff-hours per week across 3 administrators. 62% of learner traffic is mobile, against a portal that is not responsive. Legacy consent screens do not meet current PDPA requirements. The project modernises course registration and learner communications.

How to run it — step by step

#	Instruction to learners	Facilitator note
1	Take the defect data. This is the real defect log for the Contoso Training Portal Upgrade, covering sprints 1 to 7 - the same period over which Lab 20 showed CPI declining from 1.022 to 0.919. Every defect here was found after the developer marked the story done.	Check the arithmetic on one group's numbers before moving on — a wrong sign propagates into every later step.
2	Build the Pareto chart. Enter these category/count pairs exactly:	Make a group state the UNITS and the base date. Most errors here are unit errors, not maths errors.
3	Re-run Pareto by cost, and notice the disagreement. Frequency is not impact. Re-rank the same data by rework hours.	Have one group present a single row rather than the whole artifact — depth over coverage.

4	Fishbone the top category. Set the problem statement (the fish head) to exactly:	The trace is the learning, not the artifact. Ask a group to walk one line end to end aloud.
5	Drive to root cause with 5 Whys. Then work the chain. Each "why" must be answerable with evidence, not opinion.	Make a group state the UNITS and the base date. Most errors here are unit errors, not maths errors.
6	Verify the root cause before acting. A root cause you have not tested is a hypothesis. Apply the two standard tests.	Ask what the number would have to be before the decision flips. That is the exam's favourite question.
7	Design corrective and preventive actions. The distinction is exam-relevant and practically important.	Watch for groups listing only the obvious. Ask what they are deliberately leaving out, and why.
8	Answer the exam-style scenarios	Make a group state the UNITS and the base date. Most errors here are unit errors, not maths errors.

Suggested timing

Minutes	Phase
6 min	Brief the scenario, confirm the inputs and assign roles
22 min	Work the first half of the steps individually, then pool as a group
23 min	Complete the remaining steps and finalise the deliverable
9 min	Group share-back, facilitator debrief and exam-style linkage

Total: 60 minutes

Discussion & decision prompts

1. Use Pareto analysis to find the vital few defect categories driving most of the pain — how did your group decide, and what would change that decision?
2. Use a fishbone diagram to generate candidate causes across all six categories rather than the first one that comes to mind — how did your group decide, and what would change that decision?
3. Use 5 Whys to drive from a symptom to an actionable root cause — how did your group decide, and what would change that decision?
4. Your cumulative percentages reach exactly 100% and your 80% cut falls at rank 5 with 81.55% — can every member of your group defend it?
5. Where would root cause analysis with 5 whys, fishbone and pareto have saved you time or money on a real project you have worked on?

What good looks like

- Your cumulative percentages reach exactly 100% and your 80% cut falls at rank 5 with 81.55%.
- You ran the Pareto twice, on frequency and on cost, and you identified that acceptance criteria and browser compatibility rank 1 and 2 on both.
- Every fishbone category has at least two causes. An empty category means you stopped thinking, not that the category is irrelevant.
- Your root cause is a system or process failure, not a named person's shortcoming. If your chain ends in someone being careless, go further.
- Your root cause passes the removal test, and you stated honestly which defects it does *not* explain rather than over-claiming.
- Your causal chain connects the root cause to the Lab 20 CPI decline, showing that EVM detected the symptom and RCA found the cause.

- Your action list distinguishes corrective from preventive, and contains at least one action that updates an organisational process asset so a future project inherits the lesson.
- You noticed that the failure of assumption A-07 implies the other assumptions need checking, and A-05 in particular gates go-live.